

Audrey Denizot

Research Experience

- 2023.02-now Inria tenured researcher (CR), AISTroSight, Inria, Lyon, France
- 2021-2022 JSPS postdoctoral fellow, CNU, OIST, Japan. Advisor: Pr. E. De Schutter
- 2020-2021 Postdoctoral Scholar, CNU, OIST, Japan. Advisor: Pr. E. De Schutter
- 2016-2019 PhD, Beagle team, Inria, LIRIS, France. Advisors: Dr. H. Berry & Pr. H. Soula
Simulating calcium signaling in fine astrocytic processes.

Research internships in experimental & computational neuroscience

- 2016 Neurocentre Magendie, France. Advisor: Dr. A. Panatier
- 2015 Beagle team, Inria, France. Advisors: Dr. H. Berry & Pr. H. Soula
- 2013 University of Cambridge, England. Advisors: Dr. R. T. Káradóttir & Dr. H. Gautier
- 2012 Neuroscience research centre, Lyon, France. Advisors: Dr. E. Sotirakis & Pr. J. Honnorat

Education & Diplomas

- 2016-2019 PhD in computational neuroscience, LIRIS, INSA Lyon, France
- 2016 Master Biosciences (M.Sc. degree in Biology), ENS Lyon, France
- 2014-2015 Préparation à l'agrégation (biology & earth sciences teaching diploma), ENS Lyon, France
- 2012 License Biosciences (B.Sc. degree in Biology), ENS Lyon, France
- 2009-2011 Classe préparatoire scientifique, Lycée Carnot, Dijon, France

Awards & Grants

- 2025-2028 Action Exploratoire (AEx) Inria grant
- 2021-2023 Grants-in-Aid for JSPS Research Fellow (JSPS International Research Fellow) grant
- 2021-2023 JSPS Postdoctoral Standard Long-term Postdoctoral grant
 - 2021 Best Poster Award, 1st Virtual Conference of the European Society for Neurochemistry
 - 2020 2020 Trainee Professional Development Award (TPDA), Society for Neuroscience
 - 2019 Travel award for the annual CNS meeting 2019
 - 2018 LIRIS International Mobility Grant
- 2016-2019 PhD funding by the French Ministry of Superior Education (CDSN)
- 2011-2016 State Agent-Student position for excellence at ENS Lyon ("Normalien" grade)

Publications

Code: *: equal contribution

Peer-Reviewed Journals

- A. Denizot, M. Arizono, U. V. Nägerl, H. Berry, & E. De Schutter. "Control of Ca²⁺ signals by astrocyte nanoscale morphology at tripartite synapses", *Glia*, Sep. 2022, vol. 70, p. 2378-2391.

- A. Covelo, A. Badoual, A. Denizot, “Reinforcing interdisciplinary collaborations to unravel the astrocyte “Calcium Code””, J. Mol. Neurosci., May 2022.
- A. Denizot, M. Arizono, U. V. Nägerl, H. Soula, and H. Berry, “Simulation of calcium signaling in fine astrocytic processes: Effect of spatial properties on spontaneous activity”, PLOS Computational Biology, vol. 15, p. e1006795, Aug. 2019.
- K. Ceyzériat, L. Ben Haim, A. Denizot, D. Pommier, M. Matos, O. Guillemaud, M.-A. Palomares, L. Abjean, F. Petit, P. Gipchtein, M.-C. Gaillard, M. Guillermier, S. Bernier, M. Gaudin, G. Aurégan, C. Joséphine, N. Déchamps, J. Veran, V. Langlais, K. Cambon, A. P. Bemelmans, J. Baijer, G. Bonvento, M. Dhenain, J.-F. Deleuze, S. H. R. Olié, E. Brouillet, P. Hantraye, M.-A. Carrillo-de Sauvage, R. Olaso, A. Panatier, and C. Escartin, “Modulation of astrocyte reactivity improves functional deficits in mouse models of Alzheimer’s disease”, Acta Neuropathologica Communications, vol. 6, p. 104, Oct. 2018.

Preprints

- A. Denizot, M. F. Veloz Castillo, P. Puchkov, C. Cali, E. De Schutter. “The ultrastructural properties of the endoplasmic reticulum govern microdomain signaling in perisynaptic astrocytic processes”, bioRxiv, 2025, doi:10.1101/2022.02.28.482292.

Book chapters

- K. Lenk*, A. Denizot*, B. Genocchi, I. Seppälä, M. Taheri, S. Nadkarni, “Computational models of astrocyte function at glutamatergic synapses”, in: “New Technologies for Glutamate Interactions: Neurons and Glia”, pp. 229–263, Springer US, 2024.
- A. Denizot, H. Berry, and S. Venugopal, “Computational Modeling of Intracellular Calcium Signals in Astrocytes”, in: Jaeger, D., Jung, R. (Eds.), Encyclopedia of Computational Neuroscience, 2020. Springer, New York, NY, pp. 1–12.

Peer-reviewed International Conference Proceedings

- A. Ducos, A. Denizot, T. Guyet, H. Berry. “Evaluating PDE discovery methods for multiscale modeling of biological signals”. 23rd International Conference on Computational Methods in Systems Biology (CMSB), vol. 15959, Sept. 2025.
- A. Denizot, C. Cali, H. Berry, E. De Schutter. “Stochastic Spatially-Extended Simulations Predict the Effect of ER Distribution on Astrocytic Microdomain Ca²⁺ Activity”. Proceedings of the Eight Annual ACM International Conference on Nanoscale Computing and Communication (pp. 1-5), Sept. 2021.
- A. Badoual, M. Arizono, A. Denizot, M. Ducros, H. Berry, U. V. Nägerl, C. Kervrann. “Simulation of Astrocytic Calcium Dynamics in Lattice Light Sheet Microscopy Images”. IEEE 18th International Symposium on Biomedical Imaging (ISBI) (pp. 135-139). IEEE, April 2021

Presentations

Code: †: Poster, ¶: talk, ★: invited talk

- “The ultrastructural properties of the endoplasmic reticulum govern microdomain signaling in perisynaptic astrocytic processes”, “EMBO Workshop Astrocytes: From Molecules to Systems”, Venice, Italy, April 2025. †
- “Linking astrocyte nano-architecture and function: insights from computational tools”, Shape Analysis Group, McGill University, Montreal, Canada, April 2025. ★
- “Dissecting the Mechanisms Regulating Astrocyte Function at the Nanoscale with Computational Models”, Concordia University, Montreal, Canada, April 2025. ★

- "Insights into the mechanisms regulating astrocyte function at the nanoscale", "Neural computations without neurons" workshop, 22nd annual Computational and Systems Neuroscience (COSYNE) conference, Mont Tremblant, Canada, April 2025. ★
- A. Denizot, P. Puchenv, E. De Schutter. "Linking astrocyte function to cell shape: insights from computational models", French Glial Cell Club annual meeting, Sète, France, October 2024. †
- "Linking astrocyte nano-architecture and function: insights from computational tools", "Linking brain structure & function at the nanoscale: an interdisciplinary workshop", Lyon, France, September 2024. ¶
- "Linking astrocyte shape to function at the nanoscale: insights from computational approaches", Astrocyte Cafe, online, May 2024. ¶
- "Towards elucidating astrocyte function in the brain: insights from stochastic spatially-extended models", Groupe de Travail Maths-Bio, Lyon, France, May 2024. ★
- "Linking astrocyte morphology to function: insights from computational approaches", GliaLab, Oslo University, Norway, February 2024. ★
- "Dissecting astrocyte function with computational models", Simula, Numerical Analysis and Scientific Computing (SCAN) department meeting, Oslo, Norway, February 2024. ★
- A. Denizot, P. Puchenv, E. De Schutter. "Linking astrocyte function to cell shape: insights from computational models", 50th Naito conference, "Glia World - Glial Cells Governing Brain Functions", Sapporo, Japan, October 2023. †
- "Linking astrocyte nano-morphology to calcium activity at tripartite synapses", ICVS Satellite Symposium of the ISN-ESN Conference, August 2023, Braga, Portugal. ★
- Tutorial - "Modeling astrocytic calcium signaling", Annual Computational Neuroscience Meeting, Leipzig, Germany, July 2023. ★
- A. Denizot, P. Puchenv, E. De Schutter. "Computational tools to unravel mechanistic links between intracellular architecture and cell function", XVI European Meeting on Glial Cells in Health and Disease, GLIA 2023, Berlin, Germany. †
- "Dissecting the functions of astrocyte nano-architecture using voxel-based computational models", FENS Regional Meeting, May 2023, Algarve, Portugal. ★
- "Linking astrocyte morphology to calcium activity: insights from computational approaches", CEA Paris-Saclay, site de Fontenay-aux-Roses, MIRCen, April 2023. ★
- A. Denizot, M. F. Veloz Castillo, P. Puchenv, C. Cali, E. De Schutter. "The endoplasmic reticulum in fine astrocytic processes: presence, shape, distribution and effect on Ca²⁺ activity", Federation of European Neuroscience Societies Forum 2022, Paris, France, July 2022. †
- A. Denizot, M. F. Veloz Castillo, C. Cali, E. De Schutter. "Endoplasmic reticulum-plasma membrane contact sites in perisynaptic astrocytic processes: properties and effects", Neuro2022, The 45th Annual Meeting of the Japan Neuroscience Society, Okinawa, Japan, June 2022. ¶
- "The Endoplasmic Reticulum in Perisynaptic Astrocytic Processes: Properties and Effects", "All about astrocytes" symposium, organized by Pr. B. Kuhn, Okinawa Institute of Science and Technology, Japan, July 2022. ★
- "Decoding the astrocytic calcium code with computational approaches", 99th Annual Meeting of the Physiological Society of Japan, Sendai, March 2022. ★
- "Spatially-Extended Simulations Predict the Effect of ER Distribution on Astrocytic Microdomain Ca²⁺ Activity", 8th ACM International Conference on Nanoscale Computing and Communication Virtual Conference, Sept 2021. <https://bit.ly/2YXwMKq> ★
- A. Denizot, M. Arizono, V. U. Nägerl, E. De Schutter, H. Berry. "The nanoscale morphology of astrocyte branchlets governs local calcium activity", The 44th Annual Meeting of the Japan Neuroscience Society, Kobe Convention Center. July 2021. ¶

- A. Denizot, C. Cali, H. Berry, E. De Schutter. "Probing the localization of the endoplasmic reticulum in the gliapil and its effect on astrocytic calcium signals", XV European Meeting on Glial Cells in Health and Disease, GLIA 2021, July 2021, †
- "Disentangling astrocytic calcium signals: insights from spatially-extended models", Annual Computational Neuroscience Meeting, online, July 2021. <https://bit.ly/3luuDO1> ★
- "Computational approaches for simulating calcium signals in astrocytes: insights, limitations, challenges and perspectives", 1st Virtual Conference of the European Society for Neurochemistry "Future perspectives for European neurochemistry – a young scientist's conference", May 2021, ¶
- A. Denizot, C. Cali, H. Berry, E. De Schutter. "Elucidating the morphology of the endoplasmic reticulum in fine astrocyte branchlets and its effect on calcium signals", 1st Virtual Conference of the European Society for Neurochemistry "Future perspectives for European neurochemistry – a young scientist's conference", May 2021, †
- A. Denizot, C. Cali, H. Berry, E. De Schutter. "Elucidating the morphology of the endoplasmic reticulum in fine astrocyte branchlets and its effect on calcium signals", 2021 Virtual Glia Trainee Symposium, March 2021, †
- A. Denizot, C. Cali, H. Berry, E. De Schutter. "Effect of the geometry of the endoplasmic reticulum on astrocytic Ca²⁺ signals at tripartite synapses: insights from simulations in realistic 3D geometries", *SfN Global Connectome: A Virtual Event*, January 2021, †
- A. Denizot, C. Cali, W. Chen, I. Hepburn, H. Berry, E. De Schutter. "Reaction-diffusion simulations of astrocytic Ca²⁺ signaling in realistic geometries", *Annual Computational Neuroscience Meeting*, July 2020, online, <https://bit.ly/3i1bqzD> <https://bit.ly/367A27E> †
- A. Denizot, M. Arizono, W. Chen, I. Hepburn, H. Soula, V. U. Nägerl, E. De Schutter, H. Berry. "Investigating the effect of the nanoscale architecture of astrocytic processes on the propagation of calcium signals", *Annual Computational Neuroscience Meeting*, July 2019, Barcelona, Spain †
- A. Denizot, H. Soula, H. Berry, "Simulation of calcium signaling in fine astrocytic processes", *OIST Computational Neuroscience Course*, Okinawa, Japan, July 2018 †
- A. Denizot, H. Soula, H. Berry, "Simulation of calcium signaling in fine astrocytic processes: effect of spatial properties on spontaneous activity", *LyonSysBio*, Lyon, France, Nov. 2017 †¶
- A. Denizot, H. Soula, H. Berry, "Simulation of calcium signaling in fine astrocytic processes", *OIST Computational Neuroscience Course*, Okinawa, Japan, July 2017 †
- "Towards simulation of calcium signaling in fine astrocytic processes", *International Astrocyte School*, Bertinoro, Italy, March 2017 ¶
- A. Denizot, H. Soula, H. Berry, "Simulation of calcium signaling in fine astrocytic processes", *CompSysbio*, Aussois, France, March 2017 †

Conference Organization

- Sept. 2024 "Linking brain structure & function at the nanoscale: an interdisciplinary workshop".
Organizer: A. Denizot
Speakers: C. Bosch, C. Cali, C. Chirillo, E. De Schutter, A. Denizot, C. Genoud, D. Keller, C. Pape, K. Pernet-Gallay, P. Rangamani, C. Salmon
- July 2021 Annual CNS Meeting, "Computational approaches for studying astrocyte dynamics & astrocyte-neuron communication". <https://astrocytenet.org/cns2021-online-workshop/>
Organizers: B. Genocchi, A. Denizot, K. Lenk, S. Nadkarni, M. Taheri

Speakers: Y. Goda, A. Borisyuk, J. Shih, A. Scimemi, G. Yu, L. Heja, A. Pillai, R. Jolivet, M. Collard, A. Denizot, R. Refaeli, K. Lenk

May 2021 1st Virtual Conference of the ESN, "Let's join forces - Bridging the gap between experimental, computational & data sciences to disentangle astrocyte calcium activity"

Speakers: A. Covelo, A. Badoual & A. Denizot

Teaching

2016-2019 Teaching assistant: 64h/year to Master students at INSA Lyon & ENS Lyon. Main Subjects: Enzymology (M1), Cellular Biology (M2) & Neurobiology (M2)

2017 Private lessons for students preparing competitive exams to enter French Grandes Ecoles

2008-2011 Private lessons for high-school students in maths, physics and biology

Supervision

PhD students

2023.11-2026 Andr a Ducos 40% (with Dr. T. Guyet & Dr. H. Berry)

Project Partial differential equation discovery for spatio-temporal simulations in cells

2023.10-2026 Schayma Ben Marzougui 50% (with Dr. H. Berry)

Project Modeling compartmentalized second messenger networks in the retinal growth cones

2023.09-2026 Florian Dupeuble 70% (with Dr. H. Berry)

Project Biophysical modeling of neurovascular coupling at the gliovascular unit

Master students

2025.02-06 Zo  Koenig, INSA Lyon, France. 50% (with Florian Dupeuble)

Study of the geometrical properties of synapses at the gliovascular unit

23.11-24.07 Zo  Laffitte, UCBL, France. 50% (with Dr. Jan-Michael Rye)

Implementation of an open-access database of 3D cell meshes for reaction-diffusion simulations

2023.04-07 Zo  Laffitte, UCBL, France. 100%

Designing an open-access database of 3D cell meshes for reaction-diffusion simulations

2023.02-07 Mathieu Chambard, UCBL, France. 33% (with Dr. T. Guyet & Dr. H. Berry)

Multiscale Modeling with Partial Differential Equations (PDEs)

2021.09-12 Ryo Nakatani, OIST, Japan. 100%

Reaction-diffusion modeling of glutamatergic transmission at tripartite synapses

2020.10-12 Haruki Shigeta, Tohoku University, Japan. 100%

Effect of astrocyte-synapse proximity on glutamate concentration in the synaptic cleft

2017.03-07 Carlos Vivar Rios, Erasmus+ Master student. 50% (with Dr. H. Berry)

Effect of spatial constraints in realistic 3D meshes of astrocytic processes on Ca^{2+} signals

Community Service

2025 Reviewer & member of the scientific committee of the French center for the 3Rs (FC3R), which aims to reinforce the implementation of the 3Rs (Replace, Reduce, Refine) in France through education, promoting responsible and innovative research, and transparent communication.

2024 Reviewer at a scientific evaluation panel from the French National Research Agency (ANR)

- 2018-now Reviewer for Cell, PLOS Computational Biology (3), eLife, Communications Biology (2) & Psychoneuroendocrinology
- 2023-now Team representative at the committee of computer resources users (CUMI)
- 2023-2024 Member of the PhD evaluation committee of Den-Whilrex Garcia, Paris-Saclay
- 2023 Member of the 2nd year PhD evaluation committee of Aitakin Ezzati, Université Aix-Marseille
- 2020-2022 Reviewed applications to the OIST Computational Neuroscience Course 2020 & 2022
- 2021-2022 Sustainable Development Goals Advisory Group, OIST, Japan
- 2020-2022 Elected Researcher Representative of the OIST Researcher Community, OIST, Japan

Skills

Biology	Neuroscience, Biochemistry, Electrophysiology, Molecular biology, Cellular Biology
Modeling	Python, C, ODEs, Monte Carlo
Software/tools	Git, L ^A T _E X, STEPS, Blender, Trelis

Professional Development

- 2023 Training on the supervision of doctoral students ; 2 days
- 2023 Workplace First-Aid Rescuer (SST: Sauveteur Secouriste du Travail) ; 2 days
- 2023 Mental Health First Aid Rescuer (PSSM: Premiers secours en santé mentale) ; 2 days
- 2021 Neuromatch Academy: Deep Learning Course ; 3 weeks
- 2020 Leadership & Management Skills Course for Postdocs, hfp consulting ; 2 weeks
- 2017 Okinawa Computational Neuroscience Course ; 3 weeks

Outreach

2019-now **"Papier-Mâché Sciences"** association. <https://papiermachesciences.org>

Member of the board of directors & editorial committee. Goal: explain the content of scientific publications in French & outline the scientific method & publication process.

Responsibilities Author, reviewer, translator, editor, co-head of external communication

2024-2025 **"Simuler numériquement le fonctionnement de notre cerveau pour mieux le soigner"**, Université Ouverte Lyon 1, "Créer des mondes informatiques pour explorer le vivant", Villeurbanne, France. <https://uo.univ-lyon1.fr/>

2024.11 **Journée Filles et informatique : une équation lumineuse**, ENS Lyon, France ; speed-meeting with women studying in high school.

2021-2023 **"ComSciCon France"**. Proofreader & mentor for "Publithon", where PhD students have to write a scientific article accessible to a large audience. My role was to guide them through the process of scientific writing as well as following editorial guidelines.

2021.07 **"Researcher Appreciation Month"**, OIST, Japan. Co-organization, presentation of a 3min blitz talk, a poster & art pieces derived from my research.

2020.03 **"Researcher Appreciation Week"**, OIST, Japan. '200s research' talk & poster session.

2017-2020 “DéMesures” association. <https://demesures.jimdo.com/>

Aim: raising awareness of the importance of scales in science & awakening critical thinking.

Responsibilities Head of collaborations & communication, recruitment manager

- Projects
- Founder & co-manager of “Instant Recherche” (researcher interviews)
 - Founder & manager of a collaboration with the French radio “Radio Brume” & “Science pour tous” to record podcasts on topics where science and society converge
 - Co-manager of the “ArtScience” project
 - Led interactive scientific animation: “Fête de la Science” 2017, 2018 & 2019, “GeekTouch” 2017 & 2018, “A Nous de Voir” 2018, “Dans la blouse d’un chercheur” 2018
 - Creation of a photo panel on the similarities between science & art
 - Co-foundation of the “Cosmograff” project, which presented the Solar System & its scales via a guided walk in Lyon, collaboration with the Musée des Confluences & street artists
 - Creation of audio-guides, EWASS annual meeting (astronomy)
 - Interview to “Sème ta Science” 2018 to present DéMesures activities

Languages

French	Native	English	Full professional proficiency
German	Intermediate	Spanish	Basic
Japanese	Intermediate	Italian	Basic